The base model is thus used as an IIV classifier, in our reduction, by thresholding the base model's probability at a certain threshold $1/|\mathcal{E}|$. Note that such probabilities $\hat{p}(x)$ can generally be efficiently computed for base models (though efficient computation is not necessary for the lower-bounds to be meaningful).

Corollary 1. *For any training distribution p such that $p(\mathcal{V}) = 1$ and any base model $\hat{p}$,*

$$\mathrm{err} \geq 2 \cdot \mathrm{err}_{\mathrm{iiv}} - \frac{|\mathcal{V}|}{|\mathcal{E}|} - \delta,$$

for $\mathrm{err}, \mathrm{err}_{\mathrm{iiv}}$ *from Eqs.* (1) *and* (2)*, and* $\delta := |\hat{p}(\mathcal{A}) - p(\mathcal{A})|$ *for* $\mathcal{A} := \{x \in \mathcal{X} \mid \hat{p}(x) > 1/|\mathcal{E}|\}$.

Since this relationship holds for *any* base model $\hat{p}$, it immediately implies that all base models will err on inherently unlearnable IIV facts (such as the birthdays absent from the training data) where $\mathrm{err}_{\mathrm{iiv}}$ is necessarily large, and where δ and $|\mathcal{V}|/|\mathcal{E}|$ are small (e.g., for each person there are 364 times more incorrect birthday claims in $\mathcal{E}$ than correct ones in $\mathcal{V}$, plus IDK). The corollary above follows immediately as a special case of Theorem 1 which covers the more general case with prompts. Theorem 2 later uses this general result to provide lower-bounds for an intuitive special case. Theorems 3 and 4 address small $|\mathcal{E}|$, e.g., $|\mathcal{E}| = 1$ for True/False questions. The constant 2 in the above bound is relatively tight: for large $|\mathcal{E}|$ and small δ, $\mathrm{err}_{\mathrm{iiv}}$ could be near $1/2$ for unlearnable concepts while $\mathrm{err} \leq 1$. Corollary 1 also implies that $\mathrm{err}_{\mathrm{iiv}} \lesssim 1/2$.

Hallucination errors. To apply the error analysis to hallucinations, one may consider $\mathcal{E}$ to be the set of plausible generations containing (one or more) plausible falsehoods. Note that a common alternate definition of hallucinations is as *generations that are not grounded in the training data* (or prompt). Fortunately, the lower-bound above also applies to this notion because we have assumed only valid training data, i.e., a generated factual error cannot be grounded in factually correct training data.

Calibration. We now argue why $|\delta|$ is a measure of (mis)calibration that is small after pretraining. Note that *without any knowledge of the language*, one can achieve $\delta = 0$ by simply taking the uniform distribution $\hat{p}(x) = 1/|\mathcal{X}|$, and thus $\delta = 0$ does not require $p = \hat{p}$. An auditor can trivially estimate δ by comparing the fractions of responses satisfying $\hat{p}(x) > 1/|\mathcal{E}|$ versus $\hat{p}(\hat{x}) > 1/|\mathcal{E}|$ using sets of training samples $x \sim p$ and synthetic generations $\hat{x} \sim \hat{p}$. Inspired by Dawid (1982), one may think of an analogy to a weather forecaster predicting the probability of rain each day. A minimal calibration requirement would be whether their average prediction matched the average fraction of rain. One could also require these two to match on days when the forecast was $> t$ for some threshold $t \in [0, 1]$. Dawid (1982) introduced the more stringent requirement that for *every* $t \in [0, 1]$, among days on which the prediction is t it rains about a t fraction of the time.

Here is a particularly simple justification for why δ is typically small for the standard pretraining cross-entropy objective,

$$\mathcal{L}(\hat{p}) = \mathop{\mathbb{E}}_{x \sim p}[-\log \hat{p}(x)]. \tag{3}$$

Consider rescaling the probabilities of the positively-labeled examples by a factor $s > 0$ and normalizing:

$$\hat{p}_s(x) :\propto \begin{cases} s \cdot \hat{p}(x) & \text{if } \hat{p}(x) > 1/|\mathcal{E}|, \\ \hat{p}(x) & \text{if } \hat{p}(x) \leq 1/|\mathcal{E}|. \end{cases}$$

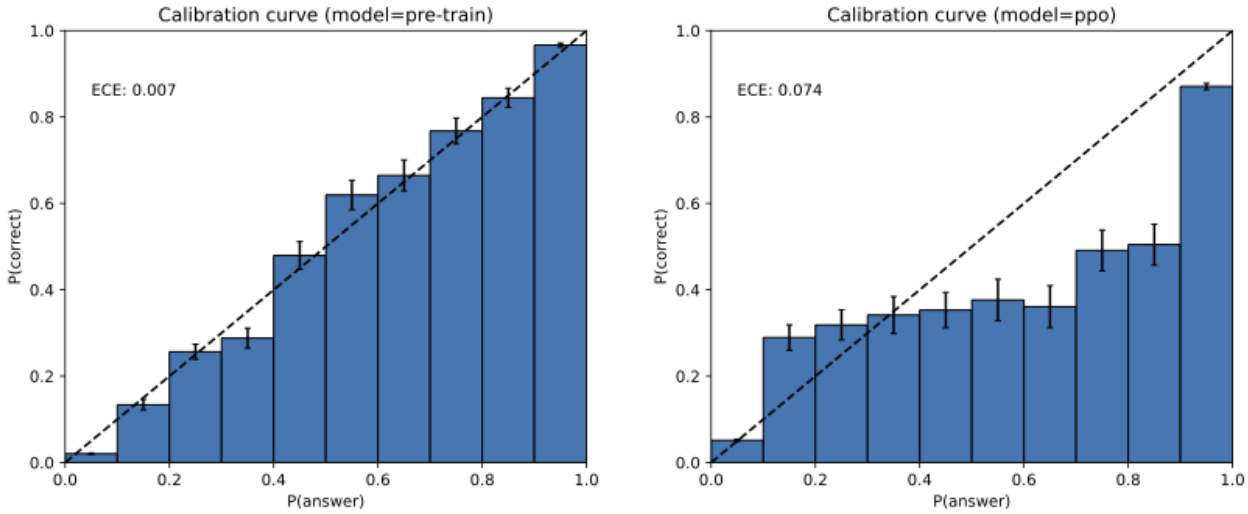

Figure 2: GPT-4 calibration histograms before (left) and after (right) reinforcement learning (OpenAI, 2023a, Figure 8, reprinted with permission). These plots are for multiple-choice queries where the plausible responses are simply A, B, C, or D. The pretrained model is well calibrated.

Then, a simple calculation shows that δ is the magnitude of the derivative of the loss with respect to the scaling factor s, evaluated at $s = 1$:

$$\delta = \left| \frac{d}{ds} \mathcal{L}(\hat{p}_s) \Big|_{s=1} \right|.$$

If $\delta \neq 0$, then rescaling by some $s \neq 1$ would reduce the loss, so the loss is not at a local minimum. For any class of language models powerful enough to approximate such simple rescaling, local optimization should yield small δ. Note that δ, being defined at a single threshold $t = 1/|\mathcal{E}|$ is weaker than notions such as Expected Calibration Error (ECE) which integrate over thresholds t.

Hallucinations are inevitable *only for base models.* Many have argued that hallucinations are inevitable (Jones, 2025; Leffer, 2024; Xu et al., 2024). However, a non-hallucinating model could be easily created, using a question-answer database and a calculator, which answers a fixed set of questions such as "What is the chemical symbol for gold?" and well-formed mathematical calculations such as "3 + 8", and otherwise outputs IDK. Moreover, the error lower-bound of Corollary 1 implies that language models which do not err must not be calibrated, i.e., δ must be large. As our derivations show, calibration—and, hence, errors—is a natural consequence of the standard cross-entropy objective. Indeed, empirical studies (Fig. 2) show that *base models* are often found to be calibrated, in contrast to post-trained models which may deviate from cross-entropy in favor of reinforcement learning.

3.2 The reduction with prompts

Henceforth, we generalize the setting of Section 3.1 to include prompts (contexts) $c \in \mathcal{C}$ drawn from a *prompt distribution* μ. Each example $x = (c, r)$ now consists of a prompt c and plausible response r. The analysis above corresponds to the special case in which μ assigns probability 1

to the empty prompt. For a given prompt $c \in \mathcal{C}$, let $\mathcal{V}_c := \{r \mid (c,r) \in \mathcal{V}\}$ be the valid responses and $\mathcal{E}_c := \{r \mid (c,r) \in \mathcal{E}\}$ be the erroneous responses. The training distribution and base model are now conditional response distributions $p(r \mid c), \hat{p}(r \mid c)$. For notational convenience, we extend these to joint distributions on $\mathcal{X}$ by $p(c,r) := \mu(c)p(r \mid c)$ and $\hat{p}(c,r) := \mu(c)\hat{p}(r \mid c)$, so that still $\mathrm{err} := \hat{p}(\mathcal{E}) = \sum_{(c,r)\in\mathcal{E}} \mu(c)\hat{p}(r \mid c)$ and $p(\mathcal{E}) = 0$.

Training distribution examples therefore correspond to valid "dialogues," as in the case of distillation (Chiang et al., 2023; Anand et al., 2023). Although assuming that the training data contain model dialogues drawn from the same prompt distribution is unrealistic, even higher error rates may be expected when the assumption fails. The IIV problem with prompts has the same target function $f(x) := +$ iff $x \in \mathcal{V}$, but the generalized distribution D selects, with equal probability either $x \sim p$ or $x = (c,r)$ for $c \sim \mu$ and uniformly random $r \in \mathcal{E}_c$. Finally, the classifier $\hat{f}(c,r)$ is now $+$ iff $\hat{p}(r \mid c) > 1/\min_c |\mathcal{E}_c|$. Corollary 1 is thus clearly a special case of,

Theorem 1. *For any training distribution p such that $p(\mathcal{V}) = 1$ and any base model $\hat{p}$,*

$$\mathrm{err} \geq 2 \cdot \mathrm{err}_{\mathrm{iiv}} - \frac{\max_c |\mathcal{V}_c|}{\min_c |\mathcal{E}_c|} - \delta,$$

where $\delta := |\hat{p}(\mathcal{A}) - p(\mathcal{A})|$ for $\mathcal{A} := \{(c,r) \in \mathcal{X} \mid \hat{p}(r \mid c) > 1/\min_c |\mathcal{E}_c|\}$.

Generalizing the rescaling $\hat{p}_s(r \mid c)$ (normalizing per prompt, still with single parameter s) again justifies a small $\delta = \left|\frac{d}{ds}\mathcal{L}(\hat{p}_s)|_{s=1}\right|$, now for $\mathcal{L}(\hat{p}) := \sum_{(c,r)\in\mathcal{X}} -\mu(c)\log\hat{p}(r \mid c)$.

3.3 Error factors for base models

Decades of research have elucidated the statistical factors contributing to misclassifications (errors in binary classification). We can leverage this prior understanding to enumerate factors behind hallucinations and other generative errors, including: statistical complexity, as in birthdays (Section 3.3.1); *poor models*, as in letter counting (Section 3.3.2); and additional factors like GIGO, as in conspiracy theories (Section 3.4).

3.3.1 Arbitrary-fact hallucinations

When there is no succinct pattern that explains the target function, there is epistemic uncertainty meaning that necessary knowledge is absent from the training data. The Vapnik-Chervonenkis dimension (Vapnik and Chervonenkis, 1971) $\mathrm{VC}(\mathcal{F})$ characterizes the worst-case number of examples needed to learn a family $\mathcal{F}$ of functions $f : \mathcal{X} \to \{-,+\}$, with high probability. Families with high $\mathrm{VC}(\mathcal{F})$ dimension may require prohibitively many samples to learn. We consider a natural special case of high VC dimension: random arbitrary facts. In particular, this section considers valid responses (other than IDK) which are random and independent across prompts.

Definition 1 (Arbitrary Facts)**.** *The following are fixed: an arbitrary prompt distribution $\mu(c)$, an* IDK *response and, for each prompt c: a response set $\mathcal{R}_c$ and a probability of answering $\alpha_c \in [0,1]$. Independently for each c, a single correct answer $a_c \in \mathcal{R}_c$ is chosen uniformly at random. Finally, $p(a_c \mid c) = \alpha_c$ and $p(\mathrm{IDK} \mid c) = 1 - \alpha_c$ for each $c \in \mathcal{C}$. Thus $\mathcal{E}_c = \mathcal{R}_c \setminus \{a_c\}$ and $\mathcal{V}_c = \{a_c, \mathrm{IDK}\}$.*

It is assumed that there is a single way to write any given fact, which can be done as in the lead birthday example where the format was specified. However, we again note that one may expect *even*